

DECODED EXPLAINED

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Critical Minerals, the foundation of strategic power

Critical minerals, the foundation of strategic power

Critical minerals have moved from the margins of resource policy to the centre of industrial strategy and security. Lithium, cobalt, nickel, graphite, copper and rare earth elements are foundational to electric vehicles, storage, renewable power, semiconductors, defence and advanced manufacturing. As decarbonisation and digitalisation accelerate, mineral security is becoming as important as oil once was. The global supply picture is concentrated. For copper, lithium, nickel, cobalt, graphite and rare earth elements, the average market share of the top three refining countries rose to 86% in 2024, from around 82% in 2020. Incremental supply is tied to a few nodes: Indonesia for nickel, and China for cobalt, graphite and rare earths. China is the leading refiner in 19 out of 20 strategic minerals with an average market share of about 70%, making minerals geopolitical, not merely commercial.

Geopolitical risk as a factor

Clean energy transitions will require minerals far beyond today's production systems. The current copper project pipeline points to a potential 30% supply shortfall by 2035. Lithium appears better supplied in the near term but rising demand is expected to drive the market into deficit by the 2030s. Rare earth demand will rise sharply as wind power, electronics and magnets expand.

Geopolitical risk is compounding these pressures. China's rare earth export controls announced in 2025 created concerns across energy, automotive, defence, aerospace, Artificial Intelligence (AI) and semiconductors. Major economies are responding quickly. The European Union (EU) Critical Raw Materials Act sets 2030 benchmarks of 10% domestic extraction, 40% processing and 25% recycling, and no more than 65% from a single country. The United States is diversifying supply chains through partnerships.

India's clean energy and manufacturing ambitions are mineral intensive. Under a net zero scenario, cumulative demand for critical energy



Vinayak Vipul

Partner, Business
Consulting,
EY-Parthenon India

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transition minerals could reach roughly 169 million tonnes by 2070, about 51% higher than under a current policy pathway. Copper demand will rise with power systems; rare earth demand will grow with wind energy and advanced manufacturing.

India has domestic potential, including reserves of cobalt (44.9 million tonnes), copper (163.9 million tonnes), graphite (211.6 million tonnes) and nickel (189 million tonnes), along with monazite deposits containing rare earth oxides. Recycling could eventually meet up to a quarter of copper and graphite demand by mid-century. Yet, reserves have not ensured supply security. India remains import dependent for lithium, cobalt and nickel, while graphite and China-dominated processing expose it to disruption as demand scales. The critical gap is processing and refining. India's position differs from the EU, the U.S. and Australia, which are building integrated supply chains through mandates, partnerships and processing investments. India is still developing foundational capability. While several minerals are processed domestically, capacity and high-purity production remain constrained. It has bulk-mineral experience but still relies on imports for high-purity critical mineral products.

This is where much value and vulnerability sit. In 2024, China accounted for over 90% of rare earths and graphite processing, nearly 75% of cobalt and 70% of lithium chemicals. India must build midstream capacity to participate meaningfully in supply chain realignment.

Structural constraints

Several constraints slow progress. Exploration remains relatively shallow, regulatory clearances can be time-consuming, private participation is limited and remote-region project economics are challenging. Processing is a larger bottleneck: India lacks some high-purity input facilities, while copper and graphite face smelting, purification and scale constraints. Recycling will play a role

but cannot substitute for primary supply in the near term because feedstock, collection and technology remain limited.

India's policy response since 2023 marks a shift. The government has identified 30 critical minerals, strengthened regulatory frameworks, and launched the National Critical Mineral Mission to support the value chain. The mission targets 1,200 domestic exploration projects by 2030-31, production of at least 15 critical minerals, and acquisition of 50 overseas mining assets by Indian companies. The Khanij Bidesh India Limited (KABIL) has secured 15,703 hectares in Argentina's Catamarca province for lithium exploration, while the 2026-27 Budget proposed rare earth corridors in Odisha, Kerala, Andhra Pradesh, and Tamil Nadu. The India-U.S. critical minerals and rare earths framework signed in May 2026 provides an additional diplomatic lever.

Mission to execution

The next phase is execution. Processing and refining must become a national industrial priority, backed by infrastructure and targeted incentives. Private capital needs better geological data, predictable approvals and risk sharing. India should also operationalise strategic stockpiles for critical minerals, accelerate applied research and development through industry partnerships, scale domestic capabilities and diversify overseas supply through trusted partners and platforms.

India energy transition, electronics, semiconductor, defence manufacturing, and advanced industrialisation all depend on secure mineral supplies. While individual policy measures are necessary, they are insufficient. A comprehensive strategy should establish mineral-specific risk thresholds, integrate recycling into supply planning, set measurable milestones, and create a coordinated institutional framework. The priority now is to convert potential into capability and reduce strategic vulnerability through sustained execution.

Critical Minerals, the foundation of strategic power

Definition: Critical minerals are metals and elements that are essential to modern technology and defence but face a high risk of supply disruption due to concentrated production. Lithium, cobalt, nickel, graphite, copper and rare earth elements are the core group.

They are foundational to electric vehicles, battery storage, renewable power, semiconductors, defence systems and advanced manufacturing. As decarbonisation and digitalisation accelerate, mineral security is becoming as strategically important as oil once was.

Critical Minerals, the foundation of strategic power

Rare Earth Minerals

They are a set of seventeen metallic elements. These include the fifteen lanthanides on the periodic table in addition to scandium and yttrium that show similar physical and chemical properties to the lanthanides.

The 17 Rare Earths are cerium (Ce), dysprosium (Dy), erbium (Er), europium (Eu), gadolinium (Gd), holmium (Ho), lanthanum (La), lutetium (Lu), neodymium (Nd), praseodymium (Pr), promethium (Pm), samarium (Sm), scandium (Sc), terbium (Tb), thulium (Tm), ytterbium (Yb), and yttrium (Y).

Critical Minerals, the foundation of strategic power

1

H

Hydrogen

$1s^1$

2

He

Helium

$1s^2$

3

Li

Lithium

$[He] 2s^1$

4

Be

Beryllium

$[He] 2s^2$

11

Na

Sodium

$[Ne] 3s^1$

12

Mg

Magnesium

$[Ne] 3s^2$

19

K

Potassium

$[Ar] 4s^1$

20

Ca

Calcium

$[Ar] 4s^2$

37

Rb

Rubidium

$[Kr] 5s^1$

38

Sr

Strontium

$[Kr] 5s^2$

55

Cs

Cesium

$[Xe] 6s^1$

56

Ba

Barium

$[Xe] 6s^2$

87

Fr

Francium

$[Rn] 7s^1$

88

Ra

Radium

$[Rn] 7s^2$

21

Sc

Scandium

$[Ar] 3d^1 4s^2$

22

Ti

Titanium

$[Ar] 3d^2 4s^2$

23

V

Vanadium

$[Ar] 3d^3 4s^2$

24

Cr

Chromium

$[Ar] 3d^5 4s^1$

25

Mn

Manganese

$[Ar] 3d^5 4s^2$

26

Fe

Iron

$[Ar] 3d^6 4s^2$

27

Co

Cobalt

$[Ar] 3d^7 4s^2$

28

Ni

Nickel

$[Ar] 3d^8 4s^2$

29

Cu

Copper

$[Ar] 3d^9 4s^1$

30

Zn

Zinc

$[Ar] 3d^10 4s^2$

31

Ga

Gallium

$[Ar] 3d^10 4s^1 4p^1$

32

Ge

Germanium

$[Ar] 3d^10 4s^2 4p^2$

33

As

Arsenic

$[Ar] 3d^10 4s^2 4p^3$

34

Se

Selenium

$[Ar] 3d^10 4s^2 4p^4$

35

Br

Bromine

$[Ar] 3d^10 4s^2 4p^5$

36

Kr

Krypton

$[Ar] 3d^10 4s^2 4p^6$

39

Y

Yttrium

$[Kr] 4d^1 5s^2$

40

Zr

Zirconium

$[Kr] 4d^2 5s^2$

41

Nb

Niobium

$[Kr] 4d^4 5s^1$

42

Mo

Molybdenum

$[Kr] 4d^5 5s^1$

43

Tc

Technetium

$[Kr] 4d^5 5s^2$

44

Ru

Ruthenium

$[Kr] 4d^7 5s^1$

45

Rh

Rhodium

$[Kr] 4d^8 5s^1$

46

Pd

Palladium

$[Kr] 4d^10$

47

Ag

Silver

$[Kr] 4d^10 5s^1$

48

Cd

Cadmium

$[Kr] 4d^10 5s^2$

49

In

Indium

$[Kr] 4d^10 5s^2 5p^1$

50

Sn

Tin

$[Kr] 4d^10 5s^2 5p^2$

51

Sb

Antimony

$[Kr] 4d^10 5s^2 5p^3$

52

Te

Tellurium

$[Kr] 4d^10 5s^2 5p^4$

53

I

Iodine

$[Kr] 4d^10 5s^2 5p^5$

54

Xe

Xenon

$[Kr] 4d^10 5s^2 5p^6$

57–71

Lanthanoids

72

Hf

Hafnium

$[Xe] 4f^14 5d^2 6s^2$

73

Ta

Tantalum

$[Xe] 4f^14 5d^3 6s^2$

74

W

Wolfram

$[Xe] 4f^14 5d^4 6s^2$

75

Re

Rhenium

$[Xe] 4f^14 5d^5 6s^2$

76

Os

Osmium

$[Xe] 4f^14 5d^6 6s^2$

77

Ir

Iridium

$[Xe] 4f^14 5d^7 6s^2$

78

Pt

Platinum

$[Xe] 4f^14 5d^9 6s^1$

79

Au

Gold

$[Xe] 4f^14 5d^10 6s^1$

80

Hg

Mercury

$[Xe] 4f^14 5d^10 6s^2$

81

Tl

Thallium

$[Xe] 4f^14 5d^10 6s^2 6p^1$

82

Pb

Lead

$[Xe] 4f^14 5d^10 6s^2 6p^2$

83

Bi

Bismuth

$[Xe] 4f^14 5d^10 6s^2 6p^3$

84

Po

Polonium

$[Xe] 4f^14 5d^10 6s^2 6p^4$

85

At

Astatine

$[Xe] 4f^14 5d^10 6s^2 6p^5$

86

Rn

Radon

$[Xe] 4f^14 5d^10 6s^2 6p^6$

89–103

Actinoids

104

Rf

Rutherfordium

$[Rn] 5f^14 6d^2 7s^2$

105

Db

Dubnium

$[Rn] 5f^14 6d^3 7s^2$

106

Sg

Seaborgium

$[Rn] 5f^14 6d^4 7s^2$

107

Bh

Bohrium

$[Rn] 5f^14 6d^5 7s^2$

108

Hs

Hassium

$[Rn] 5f^14 6d^6 7s^2$

109

Mt

Mendelevium

$[Rn] 5f^14 6d^7 7s^2$

110

Ds

Darmstadtium

$[Rn] 5f^14 6d^8 7s^2$

111

Rg

Roentgenium

$[Rn] 5f^14 6d^9 7s^2$

112

Cn

Copernicium

$[Rn] 5f^14 6d^10 7s^2$

113

Nh

Nihonium

$[Rn] 5f^14 6d^10 7s^2 7p^1$

114

Fl

Flerovium

$[Rn] 5f^14 6d^10 7s^2 7p^2$

115

Mc

Moscovium

$[Rn] 5f^14 6d^10 7s^2 7p^3$

116

Lv

Livermorium

$[Rn] 5f^14 6d^10 7s^2 7p^4$

117

Ts

Tennessee

$[Rn] 5f^14 6d^10 7s^2 7p^5$

118

Og

Oganesson

$[Rn] 5f^14 6d^10 7s^2 7p^6$

1

Group

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

57

La

Lanthanum

$[Xe] 5d^1 6s^2$

58

Ce

Cerium

$[Xe] 4f^1 5d^1 6s^2$

59

Pr

Praseodymium

$[Xe] 4f^3 6s^2$

60

Nd

Neodymium

$[Xe] 4f^4 6s^2$

61

Pm

Promethium

$[Xe] 4f^5 6s^2$

62

Sm

Samarium

$[Xe] 4f^6 6s^2$

63

Eu

Europium

$[Xe] 4f^7 6s^2$

64

Gd

Gadolinium

$[Xe] 4f^7 5d^1 6s^2$

65

Tb

Terbium

$[Xe] 4f^9 6s^2$

66

Dy

Dysprosium

$[Xe] 4f^10 6s^2$

67

Ho

Holmium

$[Xe] 4f^11 6s^2$

68

Er

Erbium

$[Xe] 4f^12 6s^2$

69

Tm

Thulium

$[Xe] 4f^13 6s^2$

70

Yb

Ytterbium

$[Xe] 4f^14 6s^2$

71

Lu

Lutetium

$[Xe] 4f^14 5d^1 6s^2$

89

Ac

Actinium

$[Rn] 6d^1 7s^2$

90

Th

Thorium

$[Rn] 6d^2 7s^2$

91

Pa

Protactinium

$[Rn] 5f^2 6d^1 7s^2$

92

U

Uranium

$[Rn] 5f^3 6d^1 7s^2$

93

Np

Neptunium

$[Rn] 5f^4 6d^1 7s^2$

94

Pu

Plutonium

$[Rn] 5f^6 7s^2$

95

Am

Americium

$[Rn] 5f^7 7s^2$

96

Cm

Curium

$[Rn] 5f^7 6d^1 7s^2$

97

Bk

Berkelium

$[Rn] 5f^9 7s^2$

98

Cf

Californium

$[Rn] 5f^10 7s^2$

99

Es

Einsteinium

$[Rn] 5f^11 7s^2$

100

Fm

Fermium

$[Rn] 5f^12 7s^2$

101

Md

Mendelevium

$[Rn] 5f^13 7s^2$

102

No

Nobelium

$[Rn] 5f^14 7s^2$

103

Lr

Lawrencium

$[Rn] 5f^14 6d^1 7s^2$

Period

1

2

3

4

5

6

7

Alkali Metal

Alkaline Earth Metal

Transition Metal

Lanthanide

Actinide

Metal

Metalloid

Nonmetal

Halogen

Noble Gas

Periodic Table of Elements

Critical Minerals, the foundation of strategic power

Refining is concentrated in a few countries: For copper, lithium, nickel, cobalt, graphite and rare earths, the average market share of the top three refining countries rose to 86% in 2024 from around 82% in 2020. Supply now depends on a handful of nodes.

China leads across nearly all strategic minerals: China is the leading refiner in 19 of 20 strategic minerals, with an average market share of about 70%.

The critical gap is processing and refining: India has bulk-mineral experience but relies on imports for high-purity critical mineral products. Capacity and high-purity production remain constrained.

Critical Minerals, the foundation of strategic power

National Critical Mineral Mission: The government has identified 30 critical minerals and launched the Mission to support the value chain, targeting 1,200 domestic exploration projects by 2030-31, production of at least 15 critical minerals, and acquisition of 50 overseas mining assets.

MMDR Act Amendment (2023): Amended the Mines and Minerals (Development and Regulation) Act to empower the central government to auction mining leases and composite licenses for 24 critical and strategic minerals (like lithium and cobalt).

Overseas acquisition through KABIL: Khanij Bidesh India Limited (KABIL), a joint venture of state-owned firms for overseas mineral assets, has secured acreage in Argentina Catamarca province for lithium exploration.

Domestic rare-earth corridors: The 2026-27 Budget proposed rare-earth corridors in Odisha, Kerala, Andhra Pradesh and Tamil Nadu.

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Q. Consider the following statements:

I. India has joined the Minerals Security Partnership as a member.

II. India is a resource-rich country in all the 30 critical minerals that it has identified.

III. The Parliament in 2023 has amended the Mines and Minerals (Development and Regulation) Act, 1957 empowering the Central Government to exclusively auction mining lease and composite license for certain critical minerals.

Which of the statements given above are correct?

[A] I and II only

[B] II and III only

[C] I and III only

[D] I, II and III

To Become Next UN Chief, Getting 'P5' Onboard Is Key

To become next UN chief, getting 'P5' onboard is key



EXPERT EXPLAINS

MANJEIV SINGH PURI

FORMER AMBASSADOR OF INDIA
TO THE UK AND NEPAL

THE UNITED Nations Security Council (UNSC) held its first "straw poll" on Thursday for choosing the next Secretary-General of the UN (UNSG), whose five-year term would start on January 1, 2022.

Although the poll's results weren't announced, three candidates from Latin America are believed to have scored well. Costa Rica's Rebeca Grynspan, an economist who heads UN Trade and Development (UNCTAD); Carolyn Birkett, Guyana's Permanent Representative to the UN; and Rafael Grossi from Argentina, who is director general of the International Atomic Energy Agency (IAEA).

Appointment process

The UNSC consists of 15 members: five permanent members (P5) — China, France, Russia, the UK, and the US — and 10 non-permanent members elected for two-year terms by the UN General Assembly.

Straw polls is a UNSC process which doesn't disclose the name of the voting member-state. The slips carry the name of the candidate and UNSC members note "encourage", "discourage", or leave them blank. While this round didn't distinctly mark the slips as belonging to the P5 or non-permanent members, the slips of the P5 will be coloured in the upcoming rounds. Drawing a "discourage" coloured slip indicates a likely P5 veto.

THE U.N. CHARTER

- The UNSC's appointment is governed by Article 17 of the UN Charter: "The Secretary-General shall be appointed by the General Assembly upon the recommendation of the Security Council."

- The Security Council's primacy in this process is clear. It forwards one name to the General Assembly for endorsement.

UNSC and no veto

The appointment of the UNSG is governed by Article 17 of the UN Charter, that "the Secretary-General shall be appointed by the General Assembly upon the recommendation of the Security Council". The primacy of the UNSC in this process is clear: it forwards just one name to the General Assembly for endorsement.

Candidates in the fray

Considering the current UNSG, Antonio Guterres, is from Portugal and his immediate two predecessors were from Asia (Ban Ki-moon of South Korea) and Africa (Kofi Annan of Ghana) respectively, the expectation is that this time, the choice will fall on someone from Latin America and perhaps a woman, as none have held the position.

Other candidates include Michelle Bachelet, former President of Chile; María Espinoza, former Foreign Minister of Ecuador; Macky Sall, former President of Senegal; and Clara Otunnu, a former UN official from Uganda. Candidates need to be endorsed by at least one member-state, though not necessarily by their own country.

The three candidates who are said to have done well in the first straw poll are from countries with good relations with the Donald Trump administration in the US. This is particularly important but the others in the P5, particularly China and Russia, may have their own views.

For instance, Grossi has had to deal with sensitive nuclear issues at the IAEA. As head of UNCTAD, Grynspan appears to have been able to positively leverage developing countries on the UNSC, while Birkett would have benefited from a certain camaraderie from fellow Ambassadors as a Permanent Representative.

Considering there's no bar on the entry of more candidates as the process unfolds, success in the end may not be with the strongest candidate in terms of multilateral and diplomatic qualifications, but one who can surmount the biggest hurdle — a certain acceptability with all of the P5.

To Become Next UN Chief, Getting 'P5' Onboard Is Key

How is the UN Secretary-General Elected?

Constitutional Provision: The UN Charter simply states that the Secretary-General is appointed by the General Assembly upon the recommendation of the Security Council.

The Power of the P5: Because the recommendation must come from the Security Council, the five permanent members (P5) (China, France, Russia, the U.K., and the US) wield decisive influence. **Any of the P5 nations can veto a candidate.**

Nominations and Public Dialogues

Joint Call for Candidates: The process officially begins when the Presidents of the General Assembly and the Security Council send a joint letter to all 193 Member States, inviting them to nominate candidates.

Vision Statements: Candidates must submit a formal curriculum vitae and a "vision statement" detailing their proposed direction for the UN.

To Become Next UN Chief, Getting 'P5' Onboard Is Key

Straw Polls:

The real power in the selection process lies within the 15-member Security Council, particularly its five Permanent Members.

The Security Council holds a series of secret ballots known as "straw polls." Members vote to "encourage," "discourage," or express "no opinion" on each candidate.

In the later rounds of straw polling, the ballots of the P5 nations are printed on different colored paper. This is crucial because any "discourage" vote from a P5 member acts as a veto, effectively eliminating that candidate from contention.

These polls continue until a single candidate emerges with sufficient support (at least nine out of 15 votes) and zero vetoes from the P5.

To Become Next UN Chief, Getting 'P5' Onboard Is Key

Formal Recommendation: Once a consensus candidate is identified, the Security Council adopts a formal resolution behind closed doors to officially recommend that individual to the General Assembly.

General Assembly Appointment: The recommended candidate's name is forwarded to the General Assembly. The Assembly votes to confirm the appointment, requiring a simple majority. However, historically, the General Assembly approves the Security Council's recommendation by consensus or acclamation without a formal vote.

NATO Enters a New Strategic Era of Transformation

NATO enters a new strategic era of transformation

For the first time since the creation of the North Atlantic Treaty Organization (NATO) in 1949, Washington is asking Europe not merely to spend more on defence but also to assume primary responsibility for its own security. This marks a fundamental shift in the transatlantic compact and signals the emergence of a new phase in the Alliance's evolution. The question is no longer whether NATO will endure, but how it is adapting to a changing strategic landscape.

America's shift

The National Security Strategy of the United States of America (November 2025) maintained NATO as America's principal transatlantic alliance while signalling a significant shift in Washington's expectations of European allies, particularly regarding burden-sharing and collective defence responsibilities. It argued that the era of the U.S. serving as Europe's primary security guarantor was drawing to a close and that prosperous European allies must shoulder greater responsibility for defending their own continent. While endorsing NATO's commitment to increase defence spending to 5% of GDP as the new benchmark for burden sharing, the strategy envisaged the U.S. remaining the Alliance's strategic enabler, nuclear guarantor and political convener rather than its sole military backbone. The objective is a stronger, more self-reliant Europe that allows Washington to devote greater strategic attention to the Indo-Pacific.

This rebalancing is already evident. Following Russia's invasion of Ukraine in 2022, U.S. troop strength in Europe rose to around 1,00,000 personnel. Since then, Washington has gradually reversed that surge. During 2025-26, it withdrew around 5,000 troops from Germany, cancelled the planned deployment of 4,000 troops to Poland, and reduced Brigade Combat Teams in Europe from four to three, bringing the U.S. military presence in Europe down to approximately 75,000-80,000 troops – close to pre-Ukraine war levels. The 36th NATO Summit in Ankara in July 2026 reinforced this transition. While reaffirming the Alliance's commitment to Article 5, members pledged €70 billion in military assistance for Ukraine in 2026, committed to sustaining comparable support in 2027, endorsed European-led development of long-range



Advitya Madan

Commanded 15 Punjab in Lebanon under the United Nations Flag, commanded 27 Sector as DIG in Churachandpur, Manipur, and was Brigadier, Operational Logistics, in HQ Western Command. He is currently an analyst on geopolitics, international and defence affairs

NATO's European shift offers India new strategic and defence opportunities

precision strike capabilities and accelerated efforts to strengthen NATO's defence industrial base. The underlying message was unmistakable: Europe must assume greater responsibility for its conventional defence as the U.S. increasingly prioritises the Indo-Pacific.

Lord Ismay, NATO's first Secretary General, famously described the Alliance's original purpose as "to keep the Russians out, the Americans in, and the Germans down". While that dictum captured the strategic logic of the Cold War, NATO today has entered a distinctly different phase that may be summed up as keeping Russia deterred, America engaged and Europe prepared.

NATO's evolution over time

The Alliance's evolution can be viewed in three phases. NATO 1.0 (1949-1991) relied overwhelmingly on American military and nuclear power to deter Soviet aggression and preserve transatlantic unity. NATO 2.0 (1991-2022) followed the collapse of the Soviet Union, expanding eastwards, incorporating former Warsaw Pact members and undertaking operations in the Balkans and Afghanistan. Yet, this period also witnessed declining European defence expenditure and growing dependence on the U.S. for intelligence, logistics, strategic mobility and high-end military capabilities, making burden sharing an increasingly contentious issue.

The Russian invasion of Ukraine has ushered in NATO's third phase, a more balanced transatlantic partnership in which Europe assumes primary responsibility for conventional defence while the U.S. continues to provide strategic enablers such as nuclear deterrence, intelligence, long-range precision strike capabilities and global power projection. This is less a story of American disengagement than one of strategic redistribution.

The scale of Europe's response was evident at the 2025 Hague Summit, where NATO members committed to increasing core defence spending to at least 3.5 % of GDP by 2035, with progress to be reviewed in 2029. Germany has doubled defence expenditure since 2023 and aims to reach the same target by 2029 while rebuilding military capability to deter Russia. Moscow, meanwhile, has reorganised its western military

posture with new headquarters, formations and bases facing NATO.

The more difficult challenge, however, is not spending but replacing the American military ecosystem. Even as European armies become stronger, they continue to depend heavily on the U.S. for intelligence, surveillance, satellite support, command and control systems, strategic airlift, missile defence, long-range precision strike capabilities and, above all, the nuclear umbrella. Washington's reduction of rapid reaction assets, including bombers, fighter aircraft, air-refuelling tankers, warships, aircraft carriers and attack submarines, together with the deferred deployment of Tomahawk cruise missiles to Germany, illustrates the extent of that dependence. Europe's central dilemma is therefore not simply building larger armed forces but developing the integrated capabilities that have long underpinned NATO's effectiveness. As Poland's Foreign Minister recently observed, Europe does not need to match American military power; it simply needs to be stronger than Russia.

New Delhi and the rebalance

For India, NATO's transformation is less about Europe than about the redistribution of American strategic attention. A Europe capable of assuming greater responsibility for its own security would enable Washington to devote more diplomatic attention, military resources and strategic investments to the Indo-Pacific, reinforcing the regional balance amid China's growing assertiveness. At the same time, a more capable Europe could emerge as an important partner for India in defence manufacturing, advanced technologies, artificial intelligence, cybersecurity, semiconductors and space.

New Delhi should therefore view these changes through the prism of strategic opportunity rather than alliance politics. Closer cooperation with both the U.S. and Europe can complement, not compromise, India's long-standing commitment to strategic autonomy. If navigated prudently, NATO's new strategic model can expand India's defence industrial partnerships, accelerate technological modernisation and strengthen its position as a leading power in an increasingly multipolar world.

NATO Enters a New Strategic Era of Transformation

The North Atlantic Treaty Organisation is a political and military alliance created in the aftermath of World War II by 12 countries from Europe and North America

Headquarters: Brussels, Belgium, while the Allied Command Operations headquarters is near Mons, Belgium.

It was formed by the signing of the Washington Treaty, which is based on Article 51 of the United Nations Charter, “which reaffirms the right of individual or collective defence.”

The principle of collective defence (an attack against one Ally is an attack against all) is enshrined in Article 5 of the North Atlantic Treaty, which commits NATO members to mutual defence.

Open door policy (Article 10): All European countries are eligible to join NATO.

NATO makes its decisions by consensus. Hence, if a new country wants to join NATO, it must be supported by all other members. Any of its member countries can veto a new member.

Funding: Every NATO country contributes to the costs of running the Alliance, based on a cost-share formula derived from Gross National Income.

Warsaw pact : It was a collective defence treaty signed in 1955 by Russia and seven other states to counter the US-led NATO. (All these seven states are now members of NATO). It was abolished in 1991.

NATO Plus : It is a grouping of NATO and five countries — Australia, New Zealand, Japan, Israel, and South Korea. The group works toward boosting global defence cooperation.

NATO Enters a New Strategic Era of Transformation

NATO Countries

- Portugal - Spain - France - Italy - Slovenia - Croatia - Montenegro - Albania - Greece - Turkey
- North Macedonia(2020)- Bulgaria - Romania - Hungary - Slovakia - Poland
- UK - Luxembourg - Netherlands - Belgium - Germany - Czech Republic
- Lithuania - Latvia - Finland
- Denmark - Finland (2023) - Iceland - Norway - Sweden (2024) (Nordic Countries)
- USA - Canada



Countries Not there : Bosnia - Belarus - Serbia - Ukraine - Moldova - Austria - Switzerland

NATO Enters a New Strategic Era of Transformation

NATO's Three Strategic Phases



2025 Hague Summit: NATO members committed to raising core defence spending to at least 3.5% of GDP by 2035.

NATO Enters a New Strategic Era of Transformation

QUESTION

GS

Hard

International Relations

Prelims 2025

Consider the following countries:

I. Austria

II. Bulgaria

III. Croatia

IV. Serbia

V. Sweden

VI. North Macedonia

How many of the above are members of the North Atlantic Treaty Organization?

Select an option to attempt

☐ A Only three

☐ B Only four

☐ C Only five

☐ D All the six

Why Must People Be Poor in a State Rich With Minerals?



BABULAL MARANDI

Why must people be poor in a state rich with minerals?

JHARKHAND POSSESSES nearly 40 per cent of India's mineral wealth, yet many of its mining districts remain among the country's least developed. For decades, coal, iron ore, bauxite and limestone have fuelled India's economic growth, while the communities living closest to these resources have continued to struggle with poor healthcare, inadequate schools, damaged roads, polluted environments and shrinking livelihood opportunities.

It was this contradiction that prompted the Centre to establish the District Mineral Foundation Trust (DMFT) in 2015. PM Narendra Modi had argued that mining-affected communities bear the greatest social and environmental costs of extraction but receive the least share of its benefits. DMFT sought to correct this imbalance by ensuring that a portion of mining revenues was reinvested in the welfare of those living in mining-affected areas. Unfortunately, that promise remains largely unfulfilled.

I recently travelled through West Singhbhum, including the Saranda mining belt, and witnessed first-hand the widening gap between the district's immense mineral wealth and the condition of its people. West Singhbhum, home to some of India's richest iron ore reserves, should have been a model of inclusive development. Between 2016 and 2026, nearly Rs 3,700 crore accumulated under its DMFT. Such resources could have transformed

Odisha's mineral resources are less than half of Jharkhand's. Yet in 2025-26, Odisha generated nearly Rs 46,000 crore in mining revenue compared with Jharkhand's Rs 22,000 crore

Saranda and surrounding mining villages through better hospitals, schools, drinking water systems, roads and livelihood programmes. Instead, what I saw were struggling local markets, shrinking economic activity, unemployed youth and communities asking a simple question: Where has the money gone?

The DMFT Rules require every district to make annual reports, budgets, approved works, project progress, governing council decisions, and beneficiary details publicly available. Yet, across Jharkhand, these disclosures are either missing, outdated or inaccessible. Mining-affected communities, for whom the fund was created, have little means of knowing how thousands of crores collected in their name have actually been spent. In Jamda, once a bustling mining township, local traders spoke of declining incomes. Several mining leases expired years ago. As mines shut down, employment disappeared. The effects spread far beyond mine workers. Transport operators lost business, roadside eateries emptied, shops struggled to survive, and young people increasingly migrated in search of work.

Cross the border into Barbil in neighbouring Odisha, barely 20 km away, and the contrast is striking. The geology is identical. The mineral deposits are the same. What differs is governance. Odisha ensured timely auctions after mining leases expired. Jharkhand hesi-

tated. The consequences are visible in the numbers. Since 2019-20, India has auctioned 434 mineral blocks. Odisha auctioned 45, Chhattisgarh 41, while Jharkhand auctioned only three, despite being the country's richest mineral-bearing state. The production gap is equally revealing. Between 2018-19 and 2024-25, Odisha's iron ore production increased from around 120 million tonnes to nearly 180 million tonnes. Jharkhand's production remained almost unchanged at approximately 23 million tonnes.

Revenue tells an even more compelling story. Odisha's mineral resources are less than half of Jharkhand's. Yet in 2025-26, Odisha generated nearly Rs 46,000 crore in mining revenue compared with Jharkhand's Rs 22,000 crore.

Delayed auctions reduce production. Lower production means lower royalty collections. Lower royalties translate into reduced contributions to DMFT. Mining-affected villages receive fewer resources precisely because the mining economy itself has been allowed to stagnate. The debate around DMFT, therefore, is not merely about financial accounting; it is about restoring trust. Before organising new investment summits or announcing ambitious industrial plans, the state must first demonstrate that it can effectively manage the resources it already possesses.

The writer is Jharkhand's first Chief Minister and currently the Leader of the Opposition in the Jharkhand Legislative Assembly

Why Must People Be Poor in a State Rich With Minerals?

● The Core Paradox

- **Jharkhand possesses nearly 40% of India's mineral wealth, yet many of its mining districts remain among the country's least developed; communities living closest to coal, iron ore, bauxite, and limestone deposits continue to struggle with poor healthcare, inadequate schools, damaged roads, and shrinking livelihoods**
- **This contradiction prompted the Centre to establish the District Mineral Foundation Trust in 2015, seeking to ensure a portion of mining revenues was reinvested in the welfare of mining-affected communities, a promise that remains largely unfulfilled**

Why Must People Be Poor in a State Rich With Minerals?

● The Odisha Comparison

- Crossing the border into Barbil in neighbouring Odisha, the contrast is striking, the geology is identical, but governance differs: Odisha ensured timely auctions after mining leases expired, while Jharkhand hesitated
- Since 2019-20, Jharkhand auctioned only three mineral blocks against Odisha's 45 and Chhattisgarh's 41, despite being the country's richest mineral-bearing state; Odisha's mineral resources are less than half of Jharkhand's, yet it generated nearly double Jharkhand's mining revenue in 2025-26

● The Vicious Cycle and the Way Forward

- Delayed auctions reduce production, lower production means lower royalty collections, and lower royalties translate into reduced DMFT contributions; mining-affected villages receive fewer resources precisely because the mining economy has been allowed to stagnate
- The debate around DMFT is not merely about financial accounting, it is about restoring trust; before organising new investment summits, the State must first demonstrate it can effectively manage the resources it already possesses

Why Must People Be Poor in a State Rich With Minerals?

Same Geology, Different Governance: Jharkhand vs Odisha

	Jharkhand	Odisha
Mineral blocks auctioned (since 2019-20)	3	45
Iron ore production (2024-25, million tonnes)	~23	~180
Mining revenue (2025-26)	₹22,000 cr	₹46,000 cr

Jharkhand holds nearly 40% of India's mineral wealth more than double Odisha's yet trails sharply on auctions, production, and revenue.

Why Must People Be Poor in a State Rich With Minerals?

DMFs are statutory bodies in India established by the state governments by notification.

They derive their legal status from **Section 9B of the Mines and Minerals (Development and Regulation) Act, 1957,** as amended on 26 March 2015 as the **Mines and Minerals (Development and Regulation) Amendment Act, 2015.**

In any district affected by mining-related operations, the **State Government shall, by notification,** establish a trust, as a non-profit body, to be called the DMF.

The Central Government has notified the rates of contribution payable by miners to the DMFs.

In the case of all mining leases executed before 12th January, 2015, miners will have to contribute an amount equal to 30% of the royalty payable by them to the DMFs.

If mining leases are granted after 12.01.2015, the rate of contribution would be 10% of the royalty payable.

Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY)

The **Ministry of Mines** launched **PMKKKY in 2015** for the **welfare of areas and people affected by mining-related operations,** using the funds generated by DMFs.

To implement various developmental and welfare projects in mining-affected areas, complementing the existing ongoing schemes of the State and Central Government

Why SC Expanded Scope of Domestic Cruelty Law to Women in Live-In Relationships

Why SC expanded scope of domestic cruelty law to women in live-in relationships

Vineet Bhalla

New Delhi, August 4

EXPANDING THE protection against domestic cruelty under criminal law, the Supreme Court has ruled that such protection, ordinarily given to married women under Section 498A of the erstwhile Indian Penal Code (IPC), extends to women in live-in relationships as well, provided the couple shares an "intent to marry".

Since the IPC has now been replaced by the Bharatiya Nyaya Sanhita (BNS), the judgment will likely apply to the corresponding BNS provision: Section 85. The judgment, delivered on Monday by a bench of Justices Sanjay Karol and N Kottiswar Singh, recognises that women facing cruelty in domestic settings cannot be denied the protection of criminal law merely because they have not undergone a formal marriage.

The legal position until now

Section 498A of the IPC penalises a hus-

band or his relatives for subjecting a woman to cruelty. Because the provision explicitly uses the word "husband", its application was traditionally restricted to legally valid marriages.

Over the years, the Supreme Court expanded this definition to cover such men who induced a woman into a marriage that is legally void or voidable — for instance, by hiding a previous subsisting marriage — holding that such men cannot claim that they were not a "husband" to escape prosecution for cruelty.

This was what had transpired in the present case as well. The accused man approached the courts to quash a Section 498A case filed by his partner, arguing that since he was already married to someone else, his relationship with the complainant was not a valid marriage. The Karnataka High Court rejected this argument, ruling that when "a man induces a woman to believe that she is lawfully married to him, and thereafter subjects her to cruelty, [he]

cannot be permitted to evade criminal responsibility on the plea that no valid marriage existed in law." This led him to appeal to the Supreme Court.

Beyond void marriages

The Supreme Court could have dismissed the appeal by relying on the established precedent regarding void marriages, but the bench took up the broader question of whether a man in a live-in relationship can be prosecuted under Section 498A.

The judgment, written by Justice Karol, adopted a "purposive interpretation" of the law, noting that since the original objective of Section 498A was to deter domestic cruelty, the law must adapt to changing social realities, observing that "cruelty, any which way, does not check at the door, whether the house it enters is that of a married woman or not". The court reasoned that if a woman faces cruelty from her partner, it makes no logical sense to offer her the protection of criminal law if she is

married, but deny it if she is cohabiting before marriage. This distinction, it said, "has no rational nexus with the object of preventing domestic violence and is as such offensive to Article 14 of the Constitution of India", which guarantees the fundamental right to equality.

Why the DV Act is not enough

During the hearings, both the petitioner and the Union government argued against expanding Section 498A, pointing out that unmarried women are already protected under the Protection of Women from Domestic Violence (DV) Act, 2005, since the Act expressly covers not just married persons but also those "related...through a relationship in the nature of marriage".

The Supreme Court rejected this argument on the basis of the fundamental difference between the two laws. The DV Act, it pointed out, is primarily a piece of civil legislation providing remedies like maintenance and protection orders. The court

said that equating it with Section 498A would be a "misstatement of law" because "the import of civil and criminal law is different" since "[o]ne has almost singularly civil and monetary consequences, whereas the other is backed by the force of criminal deterrence".

'Intent to marry' is crucial

To determine which live-in couples get this protection of criminal law, the Supreme Court borrowed the concept of a "relationship in the nature of marriage" from the DV Act. The court clarified that a "relationship in the nature of marriage" is a narrower legal subset of a general "live-in relationship". Relying on a 2013 judgment by it, the court reiterated the features of such a relationship, which include pooling financial resources, sharing domestic arrangements, having a sexual relationship, and holding themselves out to society as being akin to spouses.

Monday's judgment, however, added a caveat for the application of Section 498A:

The couple must, in addition to the above, also have the "intent to marry".

This is because, according to the court, while many relationships "may meet all the requirements to be a 'relationship in the nature of marriage'", the partners might choose never to enter matrimony. When this intent is absent, the court held, the relationship should not be equated with marriage for the purpose of invoking a penal provision and its "long list of adverse consequences".

"Those relationships that are in the nature of marriage and also have the requisite intent to marry are the closest to marriage itself, and it is those relationships that shall be entitled to the protection of [Section 498A]," the court ruled.

The judgment noted that the initial burden of proof to establish this intent will lie on the female partner seeking the law's protection. But it did not lay down specific parameters for how this internal "intent" is to be legally recognised.

Why SC Expanded Scope of Domestic Cruelty Law to Women in Live-In Relationships

The Supreme Court has expanded the scope of protection against domestic cruelty to include women in live-in relationships under certain conditions.

Originally, Section 498A of the Indian Penal Code (IPC) provided protection against cruelty to married women.

With the introduction of the Bharatiya Nyaya Sanhita (BNS), this protection now corresponds to Section 85 of the BNS.

The court recognized that women in live-in relationships should also be protected if there is an "intent to marry." : which means it applies to Women in Live - in - Relationship also

Where to apply : Judicial Activism in the matter of Gender Justice

SC Directs Centre to Frame SOPs to Curb 'Digital Arrest' Scams

SC directs Centre to frame SOPs to curb 'digital arrest' scams

The Hindu Bureau
NEW DELHI

The Supreme Court on Tuesday passed a detailed order directing the Centre, States, the Reserve Bank of India (RBI) and telecom authorities to prepare standard operating procedures (SOPs) to curb "digital arrest" scams.

The court directed them to operationalise mechanisms to redress grievances of victims, initiate criminal action through zero first information reports (FIRs) against cyber fraudsters, and restore the monetary losses of those defrauded in a time-bound manner.

The court referred to a status report submitted by the Indian Cyber Crime Coordination Centre (I4C) under the Ministry of Home Affairs, which stated that the complaints received on the National Cyber Crime Reporting Portal against digital arrests have

Complaints against digital arrests dipped from 1,23,672 in 2024 to 58,249 in 2025, and stood at 16,377 for the period ending June 30, as per the Centre's report to the court

sharply reduced from 1,23,672 in 2024 to 58,249 in 2025, and stood at 16,377 for the period ending June 30, 2026.

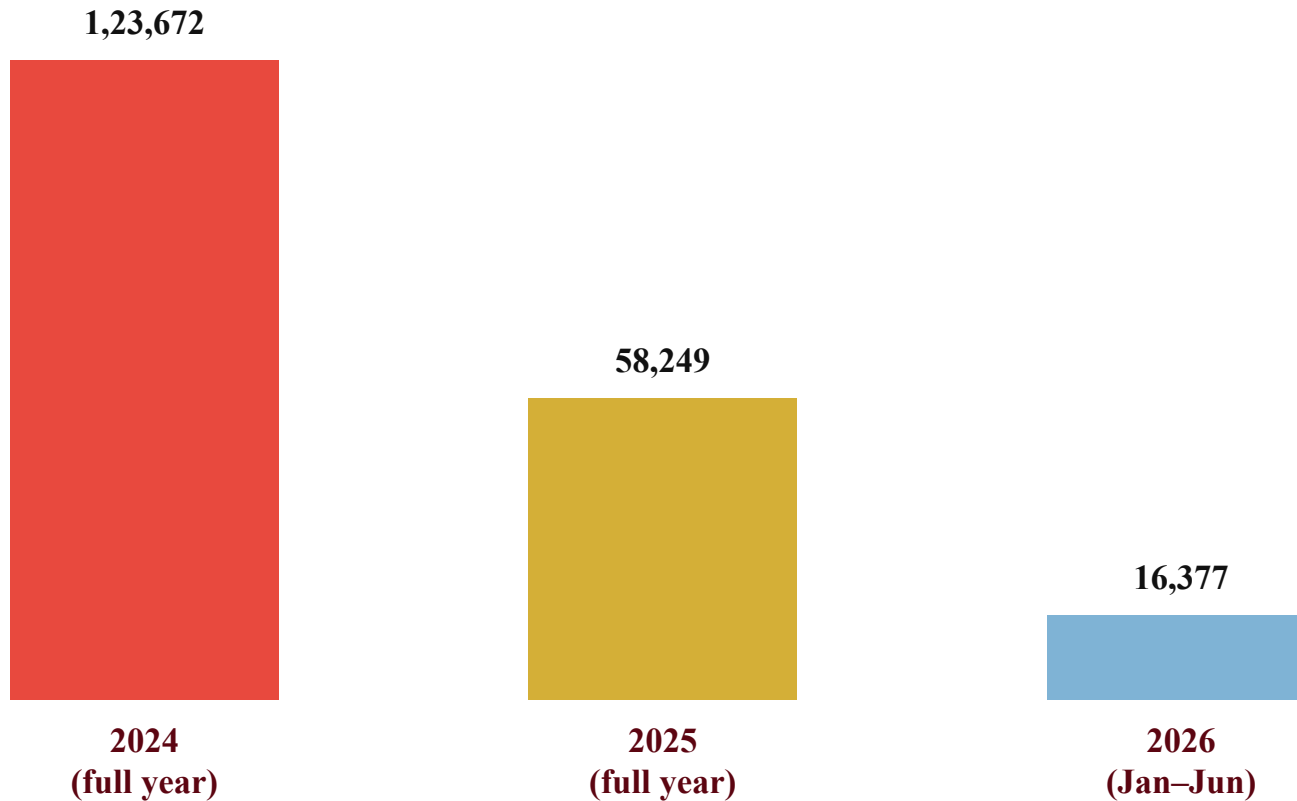
However, the court said though the statistics were "certainly encouraging", it was still essential to monitor the situation.

CONTINUED ON
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K. SWAMINATHAN SIR
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SC Directs Centre to Frame SOPs to Curb 'Digital Arrest' Scams

Digital Arrest Complaints Are Falling Sharply



Complaints on the National Cyber Crime Reporting Portal against digital arrests, per the I4C's status report to the Supreme Court.

SC Directs Centre to Frame SOPs to Curb 'Digital Arrest' Scams

Indian Cyber Crime Coordination Centre: I4C is a national-level coordination centre for addressing cybercrime issues.

It is an initiative by the Ministry of Home Affairs, approved in 2018 and dedicated to the nation in 2020. Its headquarters are in New Delhi. It provides a coordinated and comprehensive framework for law enforcement agencies (LEAs) to tackle cyber crimes.

To coordinate all activities related to the implementation of Mutual Legal Assistance Treaties (MLAT) with other countries related to cyber crimes in consultation with the concerned nodal authority in MHA.

The National Cybercrime Reporting Portal (NCRP), operationalized by the Ministry of Home Affairs, serves as a crucial platform for citizens to report cybercrimes online.

SC Directs Centre to Frame SOPs to Curb 'Digital Arrest' Scams

- **'Digital Arrest' Scams**

- A form of cyber fraud where scammers impersonate law enforcement or government officials via video or phone calls, falsely claiming the victim is under investigation or 'digital arrest', coercing them into transferring money

- **Indian Cyber Crime Coordination Centre (I4C)**

- An initiative under the Ministry of Home Affairs to provide a framework for law enforcement agencies to deal with cybercrime in a coordinated manner, operating the National Cyber Crime Reporting Portal

- **Zero FIR**

- A provision allowing a First Information Report to be filed at any police station regardless of jurisdiction, later transferred to the appropriate station, ensuring victims aren't turned away and enabling prompt action

- **National Cyber Crime Reporting Portal**

- An online platform enabling citizens to report cyber crime complaints, particularly crimes against women and children and other cyber frauds, feeding into I4C's national cybercrime data and coordination system

Thank You